

Dr. Tomer Barak | AI Advisor & Research-Infrastructure Architect

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Founder of persopy.com

Summary

PhD in computational cognition who builds the infrastructure research institutions actually run on. I advise research institutes on integrating AI into laboratory practice, and design and operate the systems that make it real — an internal platform-as-a-service, self-hosted Git and CI, an institutional LLM gateway, and the onboarding tooling that puts them in researchers' hands. My research background is in deep-learning dynamics and test-time adaptation; my engineering practice is grounded in a conclusion I reached the hard way, that fully autonomous AI agents remain unreliable for science and the value lies in grounding AI in the real record and the real workflow.

Technical Skills

Platform & Infrastructure: Rootless **Podman/Docker**, **Traefik** (TLS, forward-auth, dynamic routing), **GitLab CE** & GitLab Runner, **kaniko** CI image builds, **systemd** services, **Nginx**, LDAP SSO (**Authelia**), Let's Encrypt, Linux administration (Rocky/RHEL), NFS, XFS quotas.

AI Systems Engineering: Self-hosted **LLM** serving and gateways (**LiteLLM**, Open WebUI), Agentic Workflows (**MCP**, multi-agent topologies), agent skill/context design, retrieval over curated corpora, Distributed Job Queues (**Redis**).

Core ML & Research: **PyTorch**, Self-Supervised Learning, Contrastive Learning (InfoNCE), **Test-Time Adaptation**, Neural Dynamics (chaos theory), Python scientific stack, information theory.

Software Engineering: **Go**, **Python** (Flask, FastAPI), Full-Stack JavaScript (**Node.js**, **React**), **TypeScript**, SQL (**PostgreSQL**), REST APIs, cross-platform packaging & distribution, Git workflows.

Education

Edmond and Lily Safra Center for Brain Sciences (ELSC)

PhD in Computational Cognition & AI

Thesis: *Modeling Fluid Intelligence via Real-Time Adaptation*

Hebrew University

2017–2025

Hebrew University of Jerusalem

BSc in Physics

2012–2016

Specialisation in mathematical modelling of physical systems. Projects: degassing dynamics of vacuum metal chambers; computational simulations of chaos and fractals; electrodynamic properties of graphene. Final grade: 91/100.

Professional Experience

Edmond and Lily Safra Center for Brain Sciences (ELSC)

AI Advisor & Infrastructure Architect

Sole architect and operator of the centre's AI infrastructure — from bare servers to the tooling researchers install on their own laptops.

Hebrew University, Jerusalem

Dec 2025–Present

- **Internal Platform-as-a-Service (“Apps Hub”):** Built a Vercel-like deployment platform across three Rocky Linux servers. A researcher pushes a repo with a Dockerfile; CI builds the image with kaniko and the app goes live at a namespaced public URL, behind TLS and optional SSO, with no operator involvement.
- **Deploy orchestration & security model:** Designed a two-hop, forced-command SSH pipeline with no standing privileges — every deployment is validated for slug ownership, image provenance, and protected-namespace collisions before it runs. Apps execute as rootless Podman under hardened systemd units (all capabilities dropped, no-new-privileges, per-app network segmentation, memory/PID limits, quota-backed persistent storage).
- **Self-hosted Git, CI & public entry layer:** Deployed and operate GitLab CE with a container registry and two runner fleets (a rootless kaniko build runner, no privileged Docker daemon; and a separate deploy runner), behind a two-tier Traefik topology with Authelia LDAP forward-auth and automated certificate issuance.
- **Institutional LLM service:** Stood up local model serving with a LiteLLM gateway and a chat front-end behind LDAP group-based SSO, giving researchers institutional AI access without sending data to external providers.
- **Researcher onboarding (“Glia”):** Wrote a cross-platform **Go** bootstrapper, distributed through the **Microsoft Store**, that takes a researcher from nothing to a working AI environment on one password prompt — provisioning LLM API keys, generating SSH keys for the compute cluster and GitLab, installing a coding-agent desktop app, and configuring it against institutional models.
- **Agent context engineering:** Authored the institutional agent and a set of focused skills covering the AI services, GitLab conventions, cluster workflows, and the Apps Hub publishing contract — so that researchers' AI assistants understand the local environment rather than guessing at it.
- **Platform services & documentation:** Built and maintain the app catalogue portal, an MkDocs documentation site, and a publication-management service; wrote the deployment contract that researchers and their agents build against.

Persopy

Founder

AI personas grounded in a person's documented record — for research institutes, archives, and memorial institutions.

Jerusalem

2026–Present

- Built a pipeline that assembles an expert's complete primary corpus, reads every source in full, and distills the reusable reasoning behind it, with provenance preserved at every step.
- Personas answer *from* the sources rather than a model's recollection of them, keep a person's own words separate from later commentary, and state where the record stops instead of inventing to fill gaps. Corpora to date include Einstein, Bergson, E. T. Jaynes, and Ramón y Cajal, alongside research groups' own publication records.

Research Experience

Hebrew University, ELSC

PhD Researcher

Lab of Prof. Yonatan Loewenstein | Computational Cognition & AI

Jerusalem

2020–2025

- **Topic:** Modelled human fluid intelligence and adaptation using deep learning.
- **Test-Time Adaptation:** Adapted neural networks *during* inference, demonstrating techniques now recognised as Test-Time Adaptation.
- **Self-Supervised Learning:** Implemented contrastive learning frameworks (InfoNCE) to model predictive coding and representation dynamics without labels; used hybrid **CNN + FC** architectures to solve Raven's Progressive Matrices without prior training.
- **Dynamical Systems:** Applied physics-based analysis (phase portraits, fixed points) to reduce complex neural dynamics to tractable mathematics.
- **Human Experiments:** Designed and ran large-scale online behavioural experiments (1000+ participants via **Prolific**), with statistical analysis validating theory against human performance.

Lab of the late Prof. Naftali Tishby | Theory of Deep Learning

- Applied the **Information Bottleneck** principle to describe deep-learning dynamics.
- Simulated neural substrates including **Hopfield networks** and integrate-and-fire neurons.

Selected Earlier Projects

AI-Archive (retired)

Creator

2025–2026

A publishing platform where AI agents researched, wrote, and peer-reviewed papers under human supervision — full-stack **Node.js/React/Postgres** on **Docker**, with a custom **MCP** server and a **Redis** job queue distributing review work across machines and models. Building it is what convinced me that fully autonomous agents remain unreliable for science; that conclusion redirected my work toward grounding AI in real laboratories and real records.

Teaching & Leadership

Practical Deep Learning Course

ELSC

Co-Founder & Lead Lecturer

2019–2024

Identified a curriculum gap and co-founded the department's first accredited practical deep-learning course. Structured and maintained the syllabus over five years and four iterations — **PyTorch**, **Transformers**, optimisation dynamics (SGD → Adam), interpretability — securing accreditation and establishing a recurring practical track for theoretical researchers.

Neural Learning

ELSC

Teaching Assistant

2024

Undergraduate Research Projects

ELSC

Research Mentor

2020, 2023

Mentored three undergraduates through final research projects on deep-learning models.

Introduction to Information Processing and Learning

ELSC

Teaching Assistant

2017–2019

Selected Publications

2025: Barak, T., Loewenstein, Y. **Two pathways to resolve relational inconsistencies.** *Scientific Reports*, 15(1), 30738.

2024: Barak, T., Loewenstein, Y. **Untrained neural networks can demonstrate memorization-independent abstract reasoning.** *Scientific Reports*, 14(1), 27249.

2022: Barak, T., Loewenstein, Y. **Naive Few-Shot Learning: Uncovering the fluid intelligence of machines.** *arXiv preprint arXiv:2205.12013*.

2022: Barak, T., Loewenstein, Y. **Zero-Episode Few-Shot Contrastive Predictive Coding.** *arXiv preprint arXiv:2205.01924*.

2020: Barak, T., Avidan, Y., & Loewenstein, Y. **Naive Artificial Intelligence.** *arXiv preprint arXiv:2009.02185*.